CEFET/RJ

DAL & Comunidade

MANIPULAÇÃO DE DADOS

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Criando uma tabela básica

```r
weight <- c(60, 72, 57, 90, 95, 72)
height <- c(1.75, 1.80, 1.65, 1.90, 1.74, 1.91)
subject <- c("A", "B", "C", "D", "E", "F")
d <- data.frame(weight=weight, height=height, subject=subject)
head(d)
```

A data.frame: 6 × 3

	weight	height	subject
	<dbl>	<dbl>	<chr>
1	60	1.75	A
2	72	1.80	B
3	57	1.65	C
4	90	1.90	D
5	95	1.74	E
6	72	1.91	F

Criando uma tabela auxiliar

```r
subject <- c("A", "B", "C", "D", "E", "F")
state <- c("RJ", "SP", "MG", "RJ", "SP", "MG")
ds <- data.frame(subject=subject, state=state)
head(d)
```

A data.frame: 6 × 3

	weight	height	subject
	<dbl>	<dbl>	<chr>
1	60	1.75	A
2	72	1.80	B
3	57	1.65	C
4	90	1.90	D
5	95	1.74	E
6	72	1.91	F

Integração de dados por junção

```r
dsm <- merge(d, ds, by.x="subject", by.y="subject")
head(dsm)
```

A data.frame: 6 × 4

	subject	weight	height	state
	<chr>	<dbl>	<dbl>	<chr>
1	A	60	1.75	RJ
2	B	72	1.80	SP
3	C	57	1.65	MG
4	D	90	1.90	RJ
5	E	95	1.74	SP
6	F	72	1.91	MG

Transformações

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Conjunto de dados do pipeline são passados pelo operador |> para função seguinte no primeiro parâmetro da função de modo implícito

Pipeline: filtragem, projeção e ordenação

```
result <- dsm |>
    filter(height>1.7) |>
    select(subject, weight, height) |>
    arrange(height)
head(result)
```

A data.frame: 5 × 3

	subject	weight	height
	<chr>	<dbl>	<dbl>
1	E	95	1.74
2	A	60	1.75
3	B	72	1.80
4	D	90	1.90
5	F	72	1.91

Pipeline: agregação de dados

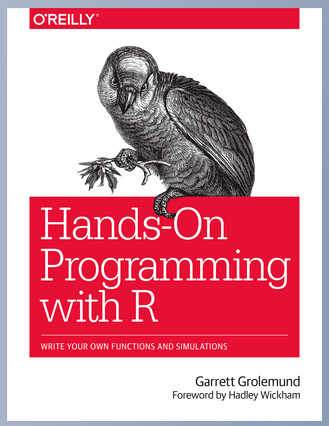
```r
result <- dsm |>
    group_by(state) |>
    summarize(count = n(), height = mean(height))
head(result)
```

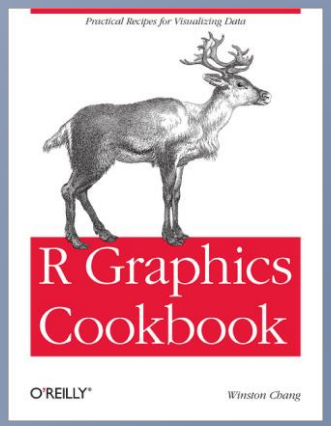
A tibble: 3 × 3

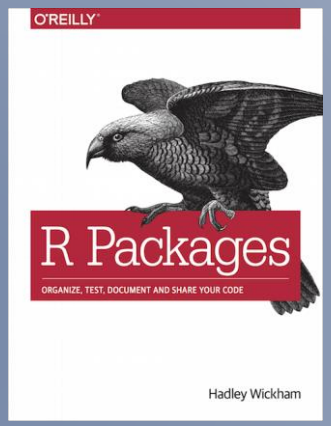
state	count	height
<chr>	<int>	<dbl>
MG	2	1.780
RJ	2	1.825
SP	2	1.770

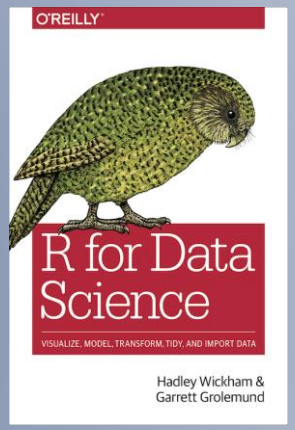
Referências

Código: https://nbviewer.org/github/eogasawara/R/blob/main/13-Manipulacao-Dados.ipynb







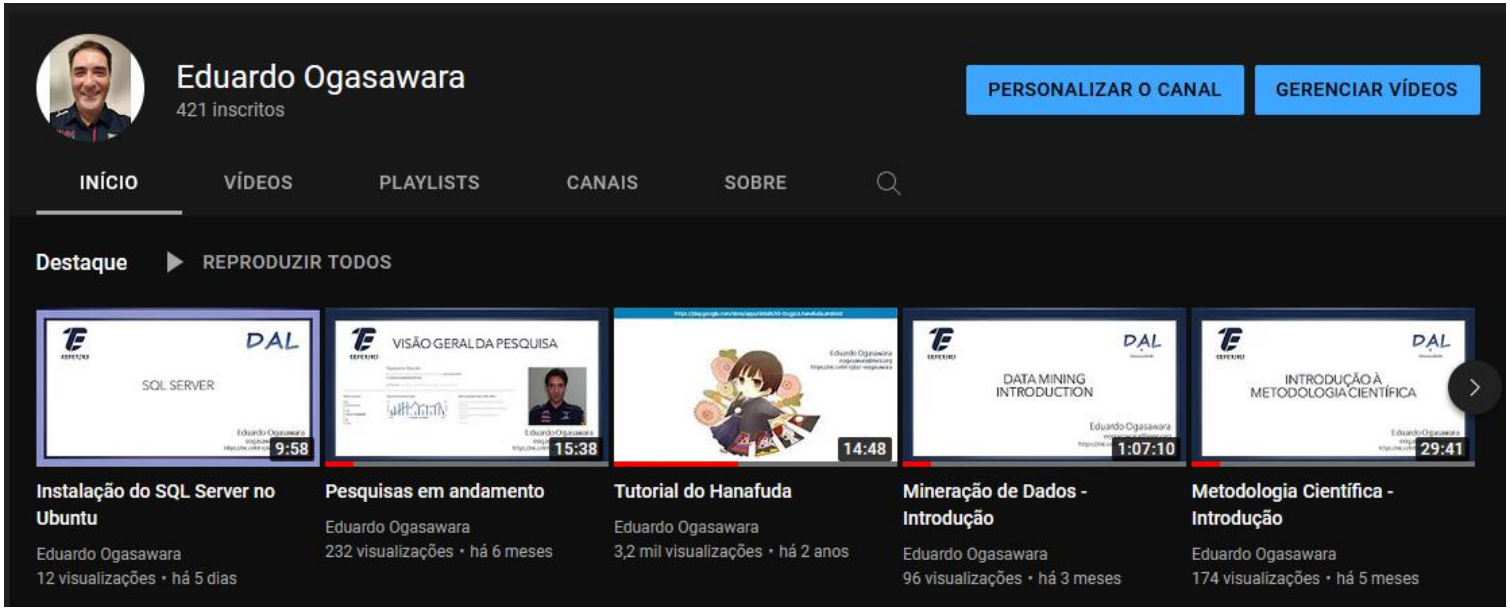


Hands-on Programming with R: https://rstudio-education.github.io/hopr/index.html
R Graphics Cookbook: https://r-graphics.org
R Packages: https://r-pkgs.org/index.html
R for Data Science: https://r4ds.had.co.nz

[1] G. Grolemund, 2014, Hands-On Programming with R: Write Your Own Functions and Simulations. O'Reilly Media, Inc.

Mantenha-se atualizado

Inscreva-se: https://eic.cefet-rj.br/~eogasawara/youtube



Slides and Jupyter notebooks: https://github.com/eogasawara/R